PO12Q - Quantitative Political Analysis: Uncovering Relationships

Week 1 - Worksheet Solutions

Dr Florian Linke

Florian.Linke@warwick.ac.uk

1.
    a. Neo-Humean Regularity focuses on the constant conjunction of events—if X and Y always happen together in time and space, a researcher relying on this path concludes a causal link exists based on that pattern alone.
    b. The mechanisms approach would be much better suited here, as it would reveal a common cause of both of these events: summer.

2. Answer: b. Counterfactual. The counterfactual logic is defined by the "what if" statement: "If the cause had not occurred, the effect would not have occurred."

3. Answer: a, c, d. A theoretical framework provides the logic ("Mechanism"), identifies potential confounders to rule out ("Alternative Explanations"), and determines the direction of the relationship ("Asymmetry). Sample size and p-values are statistical/empirical criteria, not theoretical ones.

4.
    a. A "confounding variable" (or "spurious variable").
    b. No. To satisfy this criterion, the researcher must empirically or theoretically rule out the possibility that Intelligence is the actual common cause of both Education and Income.

5. To move from a statistical association between the number of police officers and crime rates to a credible causal explanation, the researcher must address all elements of the causal framework:
    a. **Relevance within theoretical and empirical context:**
        i. **Empirical Context:** Show that observed changes in crime and policing occur in settings where other factors are reasonably stable.
        ii. **Clear Theoretical Framework:** Explain why more police should reduce crime (e.g., deterrence theory, rapid response).
        iii. **Clear Conceptualization:** Define precisely what is meant by 'more police' (number of officers, per capita, deployment patterns) and 'crime' (types, measurement).
        iv. **Exclusion of Alternative Explanations:** Control for confounding factors such as poverty, economic shifts, policing policies, or demographic changes.
    b. **Asymmetry:** Demonstrate that increasing police reduces crime, rather than high-crime areas simply attracting more police (reverse causality).

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c. **Significant Statistical Association:** Establish that the effect is not only statistically significant but substantively strong, and robust across different model specifications.

Additionally, it is helpful to specify the **mechanism** through which policing affects crime (e.g., deterrence, faster response times) to strengthen the causal explanation.

6.

a. **Temporal precedence:** Ensures that X occurs before Y, preventing reverse causation.
b. **Confounding:** A variable Z that creates a spurious association between X and Y, making the observed correlation misleading.
c. **Mediating:** A variable M that transmits the effect of X on Y along a causal pathway.
d. **Moderating (or interaction):** A variable Z that changes the strength or direction of the effect of X on Y depending on its level.
e. **Spurious correlation (or common cause):** Occurs when X and Y appear correlated only because of a shared underlying cause.
f. **Manipulation:** The principle in experimental design that allows causal inference by actively intervening on X.
g. **Mechanism:** The sequence of processes or events through which X produces Y, explaining how the causal effect occurs.

7.

a. **False.** Statistical significance indicates an association, not causality; causal inference also requires asymmetry, theoretical relevance, and the exclusion of alternative explanations.
b. **True.** In an intervening variable relationship, Z is part of the causal mechanism linking X to Y and should be explained rather than controlled away. c.**True.** Asymmetry rules out reverse causation within the same theoretical explanation, requiring that causes precede effects.
c. **True.** Multiple explanations imply that X and Z each have independent causal effects on Y, and they need not be correlated with each other.
d. **True.** The counterfactual approach evaluates causality by considering what would have happened to the outcome if the cause had not occurred.
e. **True.** An interaction effect exists when the effect of X on Y varies depending on the value of a third variable Z.

8.

a. Asymmetry → **c**
b. Alternative Explanations → **d**
c. Mechanism → **a** d.Empirical Context → **b**

9.

a. **Spurious Relationship:**
   - Z is a common cause of both X and Y.
   - Arrows: $Z \rightarrow X, Z \rightarrow Y$.
   - Remove the direct arrow between X and Y (no direct causal effect).
b. **Intervening Variable (Mechanism):**
   - X causes Y indirectly through Z.
   - Arrows: $X \rightarrow Z \rightarrow Y$.
   - Remove or leave absent the direct arrow from X to Y if the effect is fully mediated.
c. **Multiple Explanations:**
   - Both X and Z independently cause Y.
   - Arrows: $X \rightarrow Y, Z \rightarrow Y$.
   - There may or may not be an arrow between X and Z; not required for the independent effects.
d. **Interaction Effect:**
   - The effect of X on Y depends on the level of Z.

- Arrows: $X \to Y$, $Z \to Y$, and conceptually an interaction term $X \cdot Z \to Y$.
- This can be represented in diagrams as X and Z both pointing to Y with a note about the conditional effect.

e. **Common Cause Scenario:**

- This is the **spurious relationship** scenario.
- Z is the common cause making the correlation between X and Y misleading.

f. **Mediated Effect Scenario:**
   - This is the **intervening variable (mechanism)** scenario.
   - X is the primary cause, but its effect is transmitted through Z rather than operating directly on Y.

10. **(Essay Guidance)**:

    A researcher might prefer the "Manipulation" path to causality, because it involves actively intervening in the system – such as implementing a policy change in a randomized trial or pilot program. Manipulation allows the researcher to observe the direct effect of the intervention on the outcome, providing stronger evidence that the policy *produced* the change and helping to rule out spurious correlations.

    In contrast, the "Neo-Humean Regularity" approach relies on observing that the policy and outcome consistently co-occur across multiple instances or repeated observations. While regularity can suggest a pattern, it cannot on its own establish that the policy caused the outcome, since the correlation might be due to confounding factors or chance. Manipulation can demonstrate causality even with a single well-designed experiment, whereas regularity requires many repeated cases to infer a consistent pattern.

11.
    - The claim fails to sufficiently satisfy several criteria for causality. In particular, it does not adequately exclude alternative explanations, does not clearly establish asymmetry, and provides only a weakly specified mechanism.
    - The primary concern is the exclusion of alternative explanations. Neighbourhoods with more street lighting may differ systematically from others in terms of income, policing levels, community organisation, or prior crime trends. Reverse causality is also possible if areas with declining crime are more likely to receive infrastructure investment.
    - The causal claim would be strengthened by evidence from a research design that approximates manipulation, such as a natural experiment or randomized rollout of street lighting. Longitudinal data showing changes in crime following lighting installation, combined with a clearly articulated mechanism (e.g., increased visibility or deterrence), would also improve the causal argument.

12.
    - A direct causal structure would involve unemployment increasing crime through mechanisms such as financial strain or reduced opportunity costs of offending. This can be represented as $X \to Y$, where unemployment (X) directly causes crime (Y).
    - A spurious relationship could arise if a third variable, such as economic decline or deindustrialization (Z), causes both higher unemployment and higher crime. In this case, the appropriate structure is $Z \to X$ and $Z \to Y$, with no direct causal arrow between unemployment and crime.
    - An interaction structure would involve the effect of unemployment on crime depending on a third variable, such as social welfare provision or community cohesion (Z). Here, unemployment affects crime, but the strength or direction of this effect varies by the level of Z, implying an interaction between X and Z in producing Y.

13.

- The first statement represents a valid counterfactual causal claim: ‘If the policy had not been implemented, the observed improvement in outcomes would not have occurred.’ This statement directly compares the observed outcome to a hypothetical scenario in which the cause is absent.
- The second statement does not support a causal interpretation because it reverses the direction of causality. It suggests that changes in outcomes caused the policy to be adopted, rather than the policy causing the change in outcomes.
- The third statement does not establish causality because it introduces plausible alternative explanations without specifying how the effect of the policy is isolated from these other influences. While it may be empirically reasonable, it does not constitute a counterfactual claim that attributes the outcome to the policy.